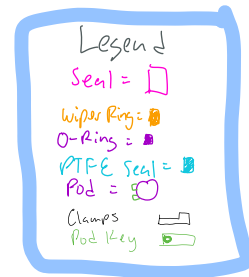
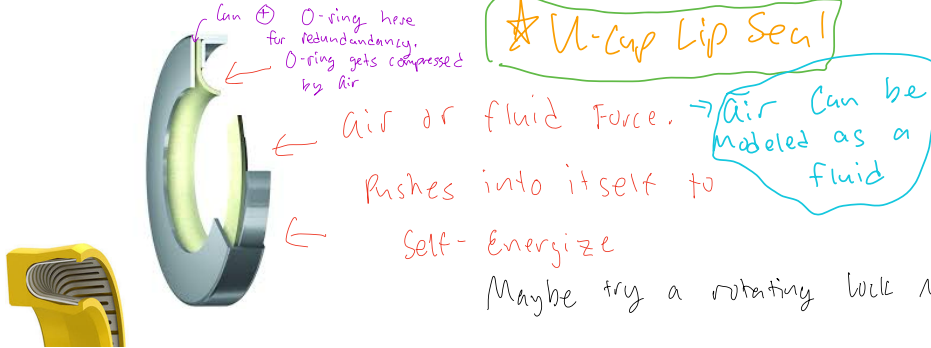
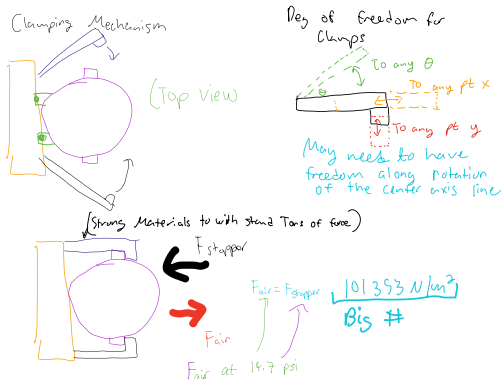
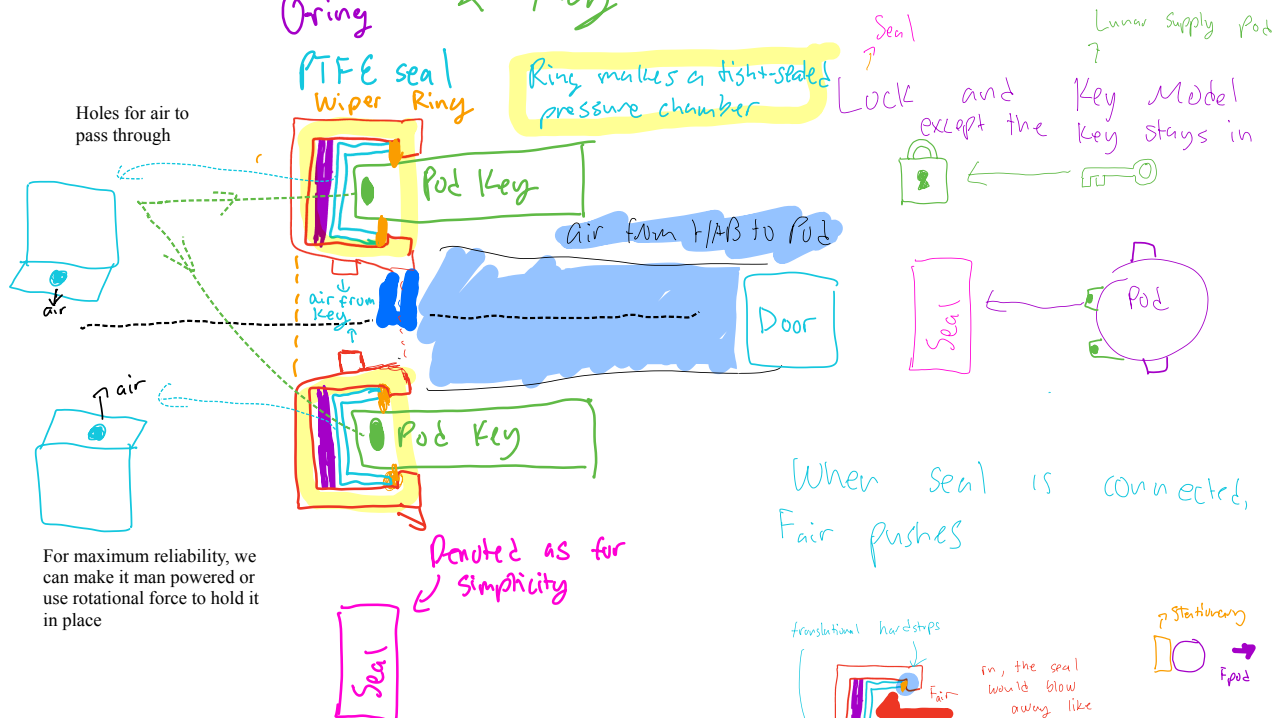


Make a seal in vacuum of space. Air pushes out into the vacuum so we are trying to keep it in.

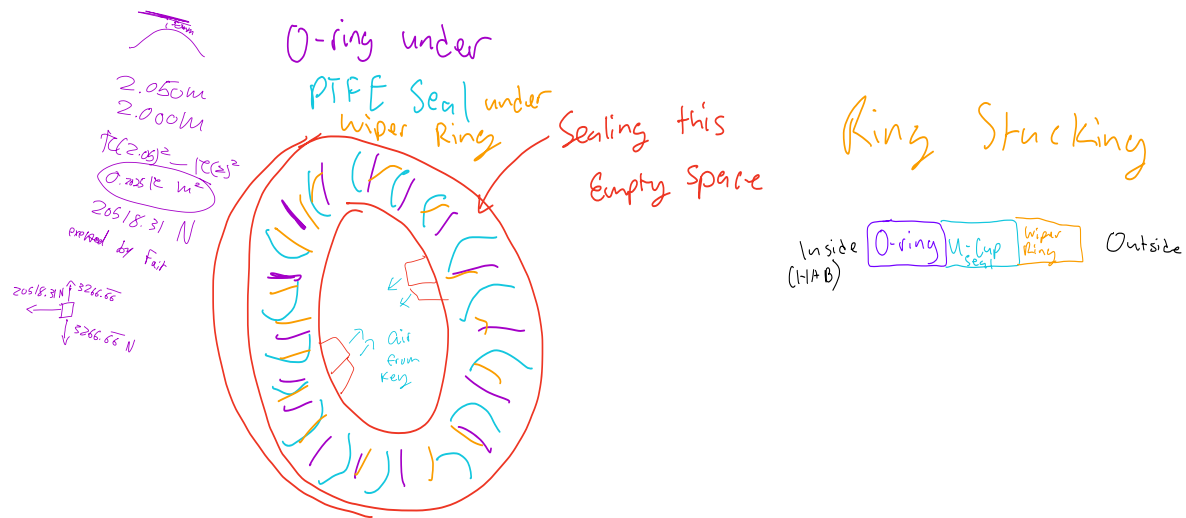
PTFE Seals utilize a self energizing concept to create the seal



Side View of Cylindrical Seal & Key



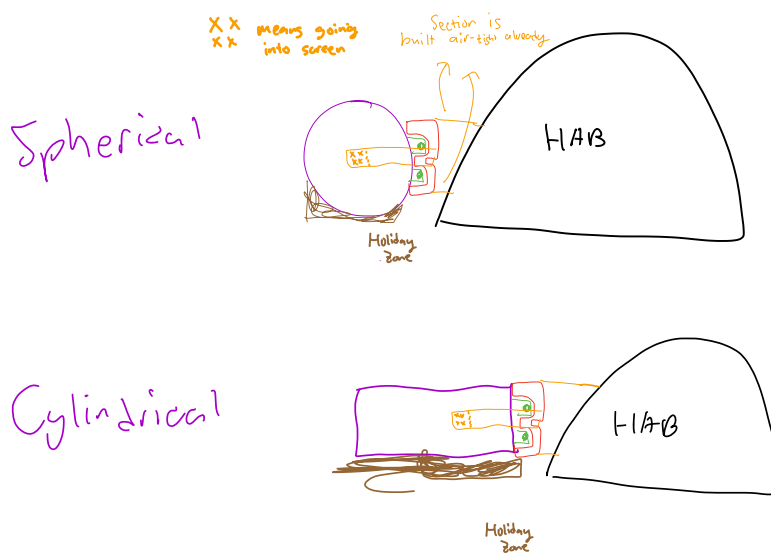
We need a reciprocating force to hold the key in place when the key is inserted into the lock. (Pod inserted into the seal) so we must have an extending translational hard stop that can be observed in the seal design.



Cylindrical pod implementation. It is just like you would do for a sphere.



This relies on a mechanical system which is infinitely more reliable than any electronic system.



Summary:

Context: lunar surface that is within a vacuum. Air is like a bullet trying to escape into the vacuum.

Goal: make an airtight seal that has dust mitigation strategies, is affordable, transportable, and temperature and abrasion resistant. (Also has long lifetime and durability)

Design

1. The Sealing Stack (Inside to Outside)

You have arranged a "Stack of 3 Rings" to handle pressure, sealing, and dust protection in a specific sequence.

- Layer 1 (Innermost / Pressurized Side): O-Ring Redundancy Located closest to the habitat atmosphere, this ring serves as the final fail-safe and redundancy for the system.
- Layer 2 (Middle): U-Cupped Lip PTFE Seal This is your primary sealing barrier. The "U-cup" geometry allows air pressure to expand the seal lips, creating a tighter seal as pressure increases.
- Layer 3 (Outermost / Vacuum Side): Wiping Ring Mechanism Located at the entrance of the seal housing, this ring physically scrapes lunar dust off the "Key" before it can enter the sealing area.

2. The Mechanical Assembly

- Lock and Key Model: The supply pod features a protruding "Key" that inserts directly into the garage's seal housing.
- Translational Stops: Physical hard stops prevent the Key from over-traveling or moving linearly once inserted.
- Clamping Mechanism: A dedicated mechanism locks the system in place, holding the Key against the translational stops to resist the force of the internal air pressure.

Pros:

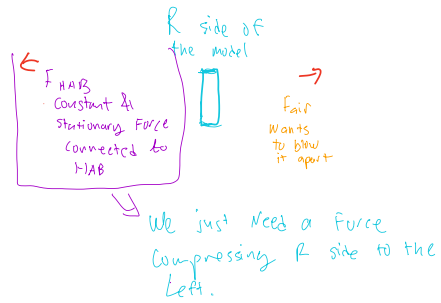
- Less material cost and light weight
- Seals effectively while having a way of dust mitigation, and the cleaning would just be replacing the wiping ring.
- Applicable to cylinder and spherical design
- Seal is self energizing

Cons:

- Requires specific pod placement and orientation. (Door facing the HAB and no freedom for rotation)
- Tight tolerance margins for pod keys and
- Requires a latch system powered by electronics or man power
- Key surface vulnerability
- Requires additional features to the supply pod
- Potential seal shrinkage at lower temps.

Next steps

- Ensure a bullet proof dust collection system
- Maybe look into the mechanisms of the clamping system?
- Material selection
- Abrasion resistance and temperature resistance
- CAD Prototyping
- Creating a working model
 - Low pressure environment
 - Way to measure pressure
 - Tennis ball
- Maybe have cut extrudes into the pods?
- Connecting the air system into HAB
- Switching directionality of the PTFE ring?
- Want the seal to connect the 2 pieces together (IMPORTANT)
 - Compressive force pushing the 2 pieces together
 - Double sided maybe???



- Might require just a simple o ring structure and maybe a PTFE seal
- Seal is only pushing against a wall which makes it pointless
- Point of the seal is to connect 2 distinct parts not prevent liquid from flowing into another sector
-

~~White edge seal~~

